

SHUBH DIWALI

Diwali Homework

Think. Explore. Write. Celebrate.

S&AI · First Year B.Tech

Chapter-wise practice questions · 10 chapters

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SECTION	A

Before you begin

Diwali is a time to rest, celebrate and recharge — but also to keep your curiosity lit. This booklet has questions from the chapters we have covered so far. Take them one at a time, enjoy the thinking, and bring your best answers back.

1. **Write in your own words.** We want to read *your* understanding, not a copied paragraph.
2. **Try first, search later.** Give every problem an honest attempt before looking anything up.
3. **Show your work.** For commands, calculations and examples, write every step you took.
4. **Keep it readable.** Write neatly and mention the chapter and question number with each answer.

How to submit

1

Write

Write your answers on your own copy (notebook). This sheet only has the questions — there is no space to answer here.

2

Photograph

Take clear, well-lit photos of your answer pages — one set of photos for each chapter.

3

Upload

Upload each chapter's photos to the public GitHub repository named **diwali-home-work**.

What's inside

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CHAPTER 1

What are Computers?

3 questions

1. Give us **10 examples from our daily life** where 3D printers are making an impact.

2. Explain the difference between machines that can be programmed and machines that cannot be programmed.

3. **Your childhood dream, in a world of computers.** There are about **28 billion computing devices** on Earth today — computers are everywhere, quietly shaping almost every field you can imagine.

   Go back to the little version of you. **What did you dream of becoming?** An astronaut, a doctor, a cricketer, a chef, an artist, a pilot, a farmer — whatever it was.
   (a) Explore how computers are used in that dream today. What do they do behind the scenes, and what has changed because of them?
   (b) Look ahead at your B.Tech journey. How can the knowledge you will gain here help you become a part of that dream — and maybe even build something new in it?
   (c) After exploring this, does that old dream feel closer than it did before? Would you like to revive it and pursue it better?

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CHAPTER 2

Hardware & Software

2 questions

1. Watch **this video on chip designing** and write a short summary of it.
   https://www.youtube.com/watch?v=dX9CGRZwD-w

2. Why did RAM get expensive in recent years?

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CHAPTER 3

CLI

7 questions

NOTE **For this chapter, write the command that does each task.** Read the documentation (man pages, **--help**) and work out the answers yourself. You may Google only after an honest 15-minute try.

1. Show all lines of a file **except the last 5**.
2. Find every file bigger than *x* MB under your home folder (or any other folder).
3. Show the **10 largest files** in **/var/log**, sorted by size.
4. Count how many **.jpg** files exist under the current folder.
5. Count the **words** in a file.
6. Show the **top 5 memory-hungry processes**.
7. Find the **last 5 commands** you ran that contain **git**.

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CHAPTER 4

Interfaces

4 questions

1. Describe 2-3 situations where a **CLI is preferred over a GUI**, and situations where a **GUI is preferred over a CLI**.
2. Create your own **custom command** that takes 2 numbers as input and outputs their sum. After that, write down everything you learned through this experience.

3 What are the different ways of passing options to a command? Comment on **GNU vs BSD style** of passing options.

4 Write a short note on the **birth of the GUI** (Xerox PARC and Macintosh).

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CHAPTER 5

Operating Systems

4 questions

1 Explain how a **single CPU** appears to run many processes at the same time.

2 “Chrome cannot directly access VS Code’s memory.” Explain why the OS gives each process its own memory space.

3 Explain how the operating system **allocates memory** to each process. What is stored in a process’s memory?

4 **The OS and your college.** Think about the last time you wrote and ran a program. You focused only on your own logic — your loops, your variables, your output. You didn't have to worry about which other programs were running, who was using the printer, how the keyboard input reached your program, or how memory was being shared. The Operating System handled all of that silently in the background. Because the OS manages processes, memory and I/O devices, you — the developer — could focus entirely on your application.

Now look around your college. Your operations team (the staff who fix the AC, repair classroom projectors, arrange chairs, manage electricity and handle room problems) does something remarkably similar. Because they handle the "background infrastructure", you — the teachers and students — can focus on what you do best: teaching and learning.

Your task: Identify **three specific responsibilities** of the operations team in your college that mirror the responsibilities of an Operating System. For each one, explain:
1. What the OS does (e.g. "The OS allocates memory to processes so they don't overwrite each other").
2. What the operations team does that is analogous (e.g. "The ops team assigns classrooms to different classes so two sections don't end up in the same room").
3. What would happen if that responsibility were removed — both in the OS and in your college.

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CHAPTER 6

Filesystem

3 questions

1 What are **environment variables** in the terminal? What is their primary purpose?

2 What is the **PATH** environment variable? Explain how the shell uses PATH to find a command.

3 What happens if PATH is empty? Why does **/bin/ls** still work? What is the **which** command used for?

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CHAPTER 7 · COMPILERS & INTERPRETERS

How Software Works?

2 questions

1. List the **advantages of compiled languages** and the **advantages of interpreted languages**.

2. "Translation always happens. The only question is *when*, and *who* does it." Explain this statement by comparing a compiler and an interpreter.

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CHAPTER 8

Number Systems

2 questions

1. Multiply $35_6 \times 15_6$ directly in base 6. Do **not** convert either number to base 10. Show both partial products and every carry.
   Which was harder — this, or 23 × 11 in base 10? Why?

2. Watch this video and write about the **impact of this video on your understanding of computers**.

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CHAPTER 9

Git & Version Control

3 questions

1. Did you participate in **Hacktoberfest**? How many git commands/concepts did you use to create and share your repository? What is your **opinion about the inclusion of this chapter** (Git's introduction) in the S&AI course?

2. Who created **Git**, when and why? Explain its origin.

3 What does **“working tree clean”** mean in the output of **git status**? Have you ever seen “working tree clean” before?

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CHAPTER 10

Computer Networks

2 questions

1 What is the **ping** command? Why does a “no reply” not always mean that a website is down?

2 A 32-bit IPv4 address space provides a theoretical maximum of 4,294,967,296 unique addresses — roughly 4.3 billion. Yet by 2020, estimates placed the number of internet-connected devices at over 20 billion. The math seems impossible: you cannot fit 20 billion unique labels into 4.3 billion slots.

Based on this information, explain how the internet is actually functioning. *Note: IPv6 is not the answer.*

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